

Scaling OCP for Supercomputing: HPE Cray GX5000

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50 years after Cray-1, HPE Cray SC GX5000 is next

1975* — Cray-1 supercomputer



1985* — Cray-2 supercomputer



2018* — HPE Cray SC EX4000



2025* — HPE Cray SC GX5000



*Year of public announcement

Convergence of HPC and AI Infrastructure

- Power and Cooling requirements pushing beyond current Exascale platforms
- Increasingly heterogeneous and converged HPC & AI workflows
- Proliferation of workflow optimized accelerators and interconnects
- Leverage and build upon existing OCP standards and industry momentum
- And more…

HPE Cray SC GX5000 at a glance



- 400KVA+ per cabinet
- Near 100% Direct Liquid Cooled (DLC) and 40C facility water support
- Up to 144 nodes (73,728 cores) per Cabinet
- Mix CPU and accelerator blades within cabinet
- HPE Slingshot optimized or industry Interconnects
- Scales to 100's of cabinets

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H L R S
High-Performance Computing Center Stuttgart



lrz Leibniz-Rechenzentrum der Bayerischen Akademie der Wissenschaften



HPE Cray SC GX5000 - Cabinet

Cabinet Dimensions:

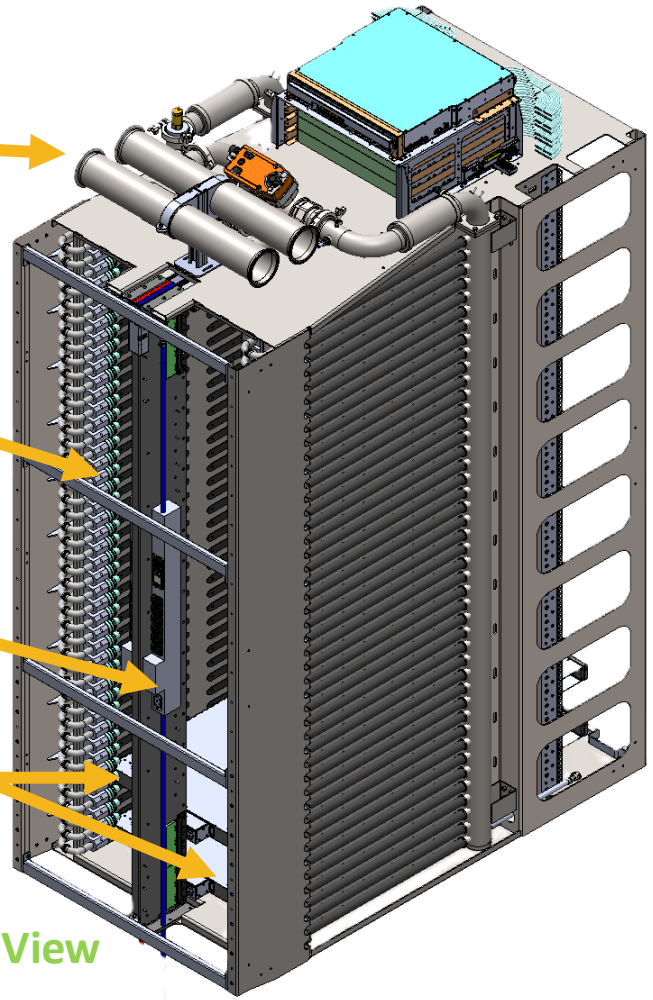
Width:	894mm / 35.43”
Depth:	1,642mm / 64.45”[1]
Height:	2,426mm / 95.5”[2]
Footprint:	900mm x 1200mm[3]
Weight:	4,535kgs/10,000lbs

High flow 2.5” fluid manifold connections with cabinet isolation valves

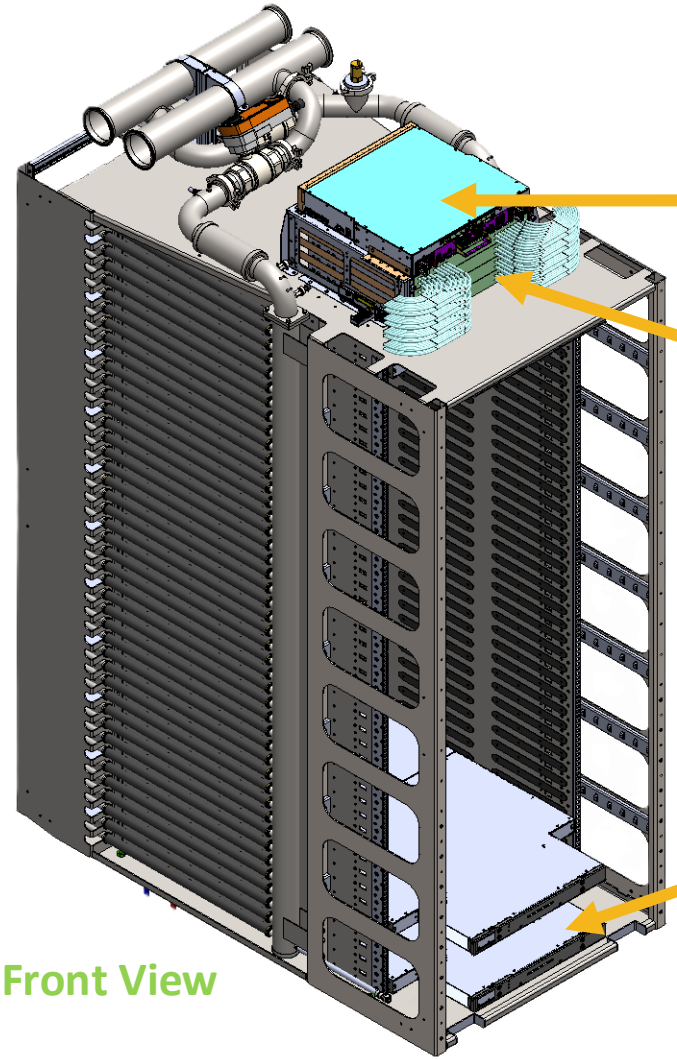
Secondary Fluid Manifolds and high flow quick disconnects (CGD12)

50V (8000A) DLC Bus Bar

In-rack scale-up AI networking ready



Rear View



Front View

Rack Management Controller

Redundant, air-gapped Ethernet Management Switches

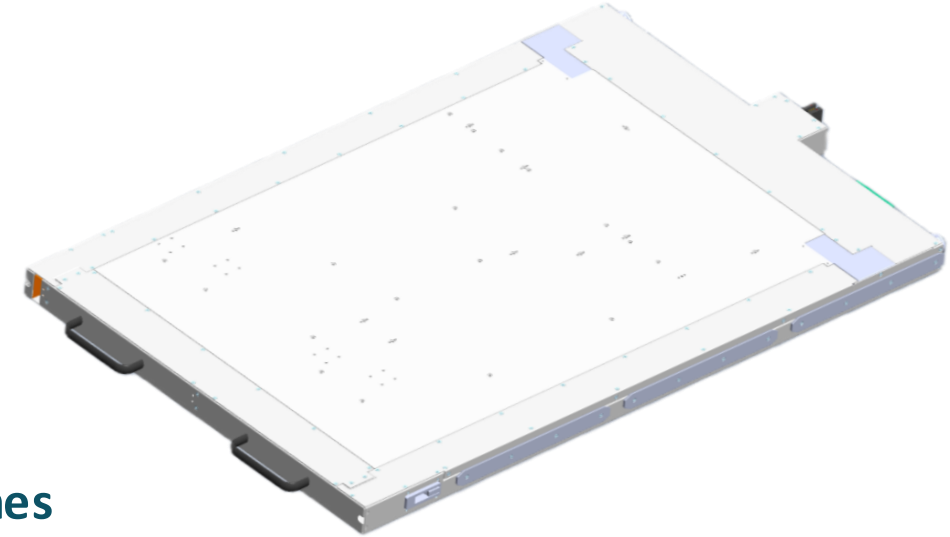
44 OU space for compute, high-speed networking, and power

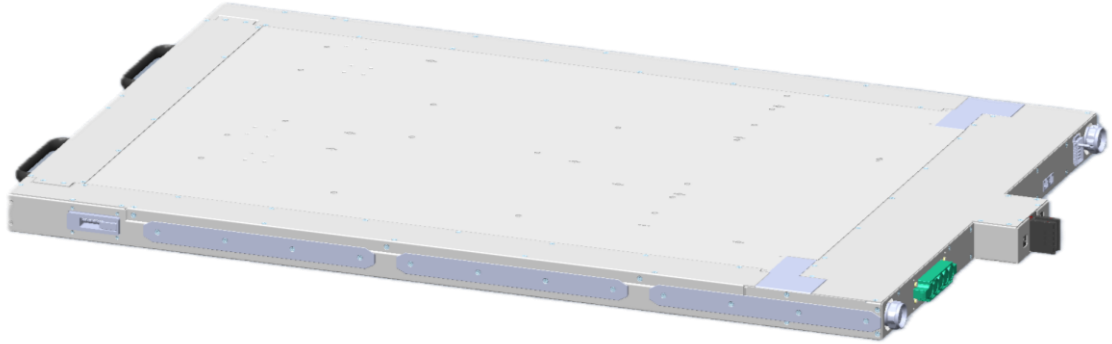
DLC power shelves (not shown)

[1] with doors, rack extensions, power feeds
[2] with rack management controller and switches
[3] without doors or rack extensions

HPE Cray SC GX5000 – Power

- **100% fan-less, direct liquid cooled power shelf**
  - Up to 66kW output capacity per 1 OU
  - Fail in place (2x 33kW rectifiers), service in place (entire shelf)
  - 97%+ efficiency at >30% load
  - Integrated energy storage for GPU transients
  - 100A x 480V input or 125A x 400V; 4-pin (three phase + PE)
- **Up to 25kW per OU for compute blades and network switches**
- **Future scalability beyond 400kW including 800VDC**

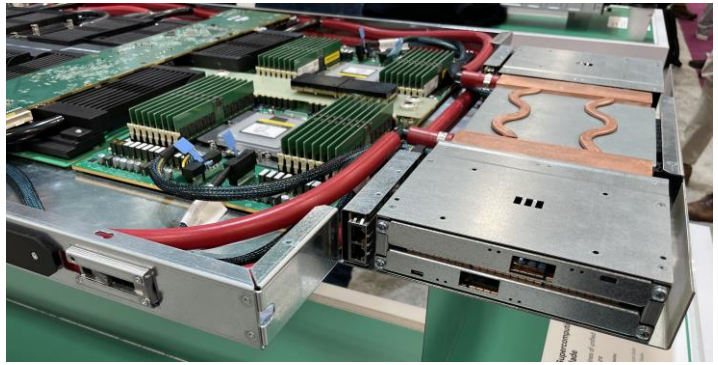


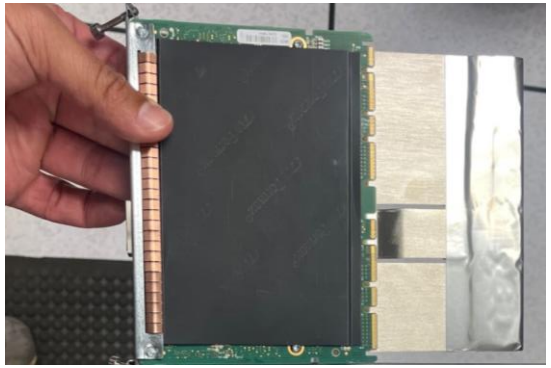


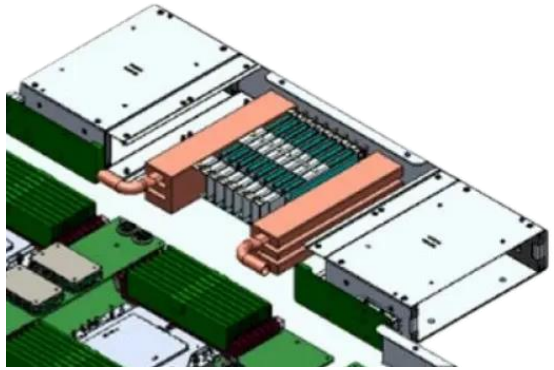
HPE Cray SC GX5000 – Cooling

- **High flow capacity to support 40C facility water and increasing GPU power**
  - Large capacity manifolds with 2.5” top/bottom feed support
  - CGD12 quick disconnects per OU enable 20kW+ cooling at 1.5 lpm/kW
  - Ready for 800VDC and higher rack power
- **Narrow edge E1.S cooling specification co-contributor (SNIA, OCP?)**
- **Dry disconnect cooling for OCP NIC 3.0 single and double wide adapters**

HPE was leading contributor for TDSFF double-wide OCP NIC 3.0 specification (version 1.5)



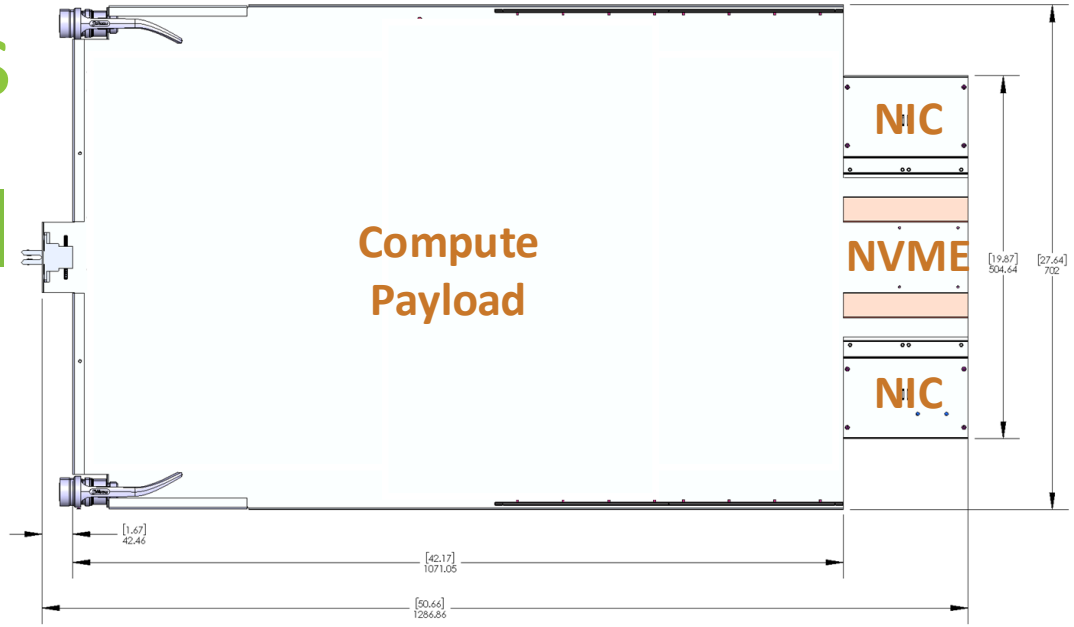




Source: https://blog.glennklockwood.com/2025/12/sc25-recap.html

HPE Cray SC GX5000 – Blades

Feature	Description
Blade Name	GX250
Node Type	Dual socket AMD Venice
Nodes per Blade	4 x 2P nodes
Processor	2 x AMD Venice SP7, 500W/600W, <=256 cores per CPU
Memory	16 x DIMM per CPU, 1 DIMM per channel MRDIMMS 12,800MT/s DDR5 DIMMS 8,000MT/s
Management Network	1Gb Ethernet (Building Management System) 10 Gb Ethernet (Node Management)
High speed Fabric	4 or 8 x HPE Slingshot-400/800 endpoints per blade
Storage Options	2 x E1.s PCIe Gen5+ x4 per node
Density	Up to 512 processors per Scalable Unit (1 Slingshot NIC per CPU) Up to 1024 processors per Scalable Unit (1 Slingshot NIC per node)





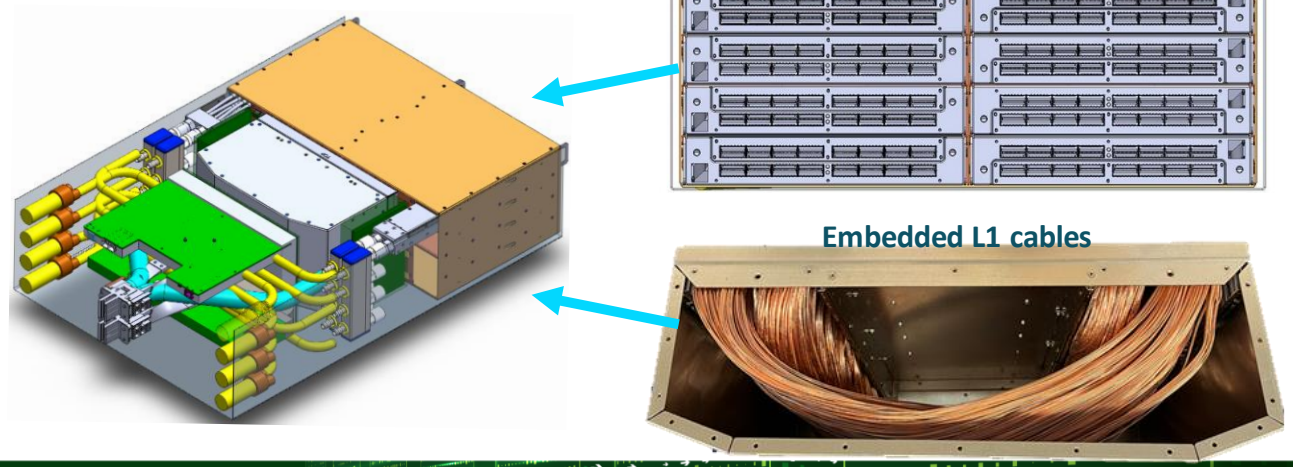
HPE Cray SC GX5000 – DLC Networking

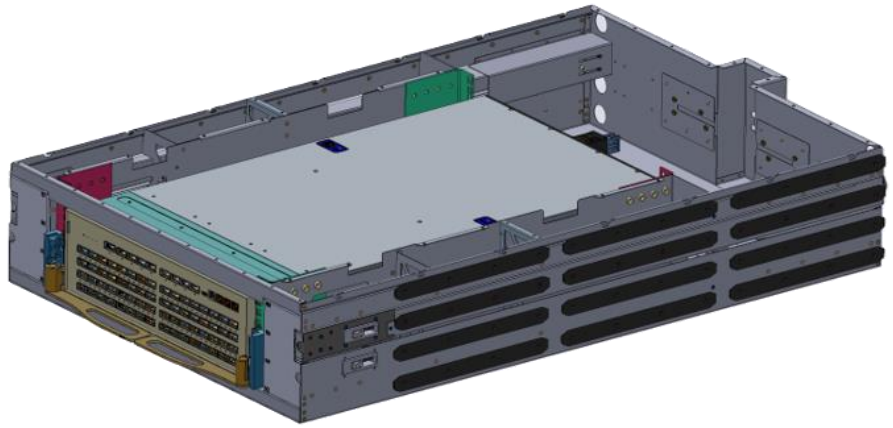
2nd and 3rd Generation HPE Slingshot

- Superior architecture for low-latency lossless Ethernet
- “Connectionless” RDMA – pioneers path to UET
- GX form-factor enhances flexibility and retains system price-performance optimizations
  - New flexibility for network and compute topologies
  - NIC flexibility – can accommodate GPU embedded Ethernet
  - Standardized NIC mezzanine FF (OCP double-wide)
  - Embedded cabling - speeds deployment, reduces power, reduces cost

Third Party Support / OCP Blades

- Enables off-the-shelf OCP components (switches, compute blades, power components) in GX5000
- Cooling – convert QD sizes from GX5000 to MGX/ORv3
- Power – bridge to busbar + isolation control
- NVIDIA Quantum-X800 InfiniBand (announced at GTC 2026)





GX5000 OCP/MGX Adapter Kit

Call to Action

- Problems to solve
  - High node density requirements while still leveraging industry momentum with OCP technology
  - Lack of Plug and play interoperability of serviceable, liquid cooled, compute, power, networking subsystems and IO modules
  - Bringing liquid cooling support to OCP NICs without losing front panel or slot density
  - Lack of common liquid cooled power shelf capabilities across the input power spectrum (800VDC / legacy 415, 480VAC)
  - Lack of standards for liquid quick disconnect size and flow requirements for continued GPU, rack power increases
- Increase OCP focus in the HPC and SuperComputer space
  - Work with rack and power WG to consider adding an HPC subteam to involve HPE GX5000 team
  - Explore how GX5000 infrastructure can be shared with OCP community
- How to get involved in the project/sub-project community
  - Collaborate on standardization of modular, liquid cooled, subsystems that scale beyond ORV3

Thank You!

